3. Further complex numbers

What students need to learn:

Euler's relation $e^{i\theta} = \cos\theta + i\sin\theta$.	Students should be familiar with $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$.
De Moivre's theorem and its application to trigonometric identities and to roots of a complex number.	To include finding $\cos n\theta$ and $\sin m\theta$ in terms of powers of $\sin\theta$ and $\cos\theta$ and also powers of $\sin\theta$ and $\cos\theta$ in terms of multiple angles. Students should be able to prove De Moivre's theorem for any integer n.
Loci and regions in the Argand diagram.	Loci such as $\lvert z - a\rvert = b$, $\lvert z - a\rvert = k\lvert z - b\rvert$, $\arg(z - a) = \beta$, $\arg\frac{z-a}{z-b} = \beta$ and regions such as $\lvert z - a\rvert \leqslant \lvert z - b\rvert$, $\lvert z - a\rvert \leqslant b$.
Elementary transformations from the z-plane to the w-plane.	Transformations such as $w = z^2$ and $w = \frac{az+b}{cz+d}$, where $a, b, c, d \in \mathbb{C}$, may be set.